

CHARITON, IA, USA

WMO: 725469

Lat: 41.019N

Lon: 93.359W

Elev: 1050

StdP: 14.15

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04913

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.5	1.5	-10.9	3.2	-2.5	-6.3	4.1	4.3	28.7	21.9	26.6	21.0	8.1	310	0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	92.9	77.7	90.4	76.9	87.9	75.6	80.5	89.5	78.7	87.3	77.1	85.2	10.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.5	153.9	87.4	76.5	144.0	85.2	74.3	133.1	82.9	45.2	89.9	43.0	87.8	41.3	85.0	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.9	21.6	19.1	-10.7	97.2	7.8	3.4	-16.3	99.6	-20.9	101.6	-25.2	103.5	-30.9	106.0		
			-11.5	82.8	6.6	1.6	-16.2	84.0	-20.1	84.9	-23.9	85.8	-28.7	86.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.9	25.3	28.5	39.7	52.2	62.5	72.0	76.4	73.5	66.6	53.9	41.3	29.0
	DBStd	20.30	12.35	12.48	11.54	9.61	8.16	6.40	5.83	5.49	8.03	9.57	10.67	11.59
	HDD50	2842	767	605	359	85	5	0	0	0	1	68	298	654
	HDD65	5868	1231	1021	786	394	146	15	2	4	76	363	713	1117
	CDD50	3523	0	4	39	151	394	659	819	728	500	189	37	3
	CDD65	1073	0	0	2	10	70	224	356	267	124	19	1	0
	CDH74	9810	0	0	10	132	592	2010	3548	2270	1086	157	5	0
	CDH80	3579	0	0	1	25	151	719	1488	805	360	30	0	0
Wind	WSAvg	9.1	10.1	10.1	10.8	11.1	9.7	8.1	6.8	6.3	7.5	9.0	10.0	9.7
Precipitation	PrecAvg	36.1	1.0	1.3	2.0	3.9	4.9	5.4	4.3	4.1	3.5	2.4	1.9	1.3
	PrecMax	52.9	2.2	2.4	5.4	8.0	9.6	13.0	12.4	10.0	7.2	6.1	4.9	4.7
	PrecMin	23.4	0.6	0.2	0.0	1.6	1.2	1.0	1.0	0.1	1.7	0.3	0.0	0.1
	PrecStd	7.2	0.4	0.6	1.3	1.5	2.0	2.9	2.7	2.6	1.6	1.5	1.2	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.2	64.2	74.7	83.7	89.5	93.0	98.9	94.9	91.4	84.1	73.2	61.0
		MCWB	47.7	53.3	62.3	67.8	75.2	78.2	77.2	76.7	74.7	69.5	60.3	54.7
	2%	DB	51.8	57.1	68.4	77.5	84.0	90.3	94.5	91.0	88.2	78.9	67.8	54.3
		MCWB	44.8	49.2	57.4	62.0	70.8	76.3	79.2	77.5	73.8	64.9	57.4	48.3
	5%	DB	45.8	51.7	63.4	72.8	80.8	87.9	90.9	87.9	84.1	73.4	62.9	49.5
		MCWB	40.7	45.3	54.6	60.1	67.8	75.0	77.8	76.4	72.5	62.3	53.7	45.3
	10%	DB	41.1	45.3	57.4	68.4	76.9	84.1	88.2	84.1	80.9	69.9	57.1	44.8
		MCWB	36.6	40.3	49.4	57.3	65.8	73.2	76.7	74.4	69.9	60.8	49.2	40.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.4	57.1	64.2	69.3	76.5	80.4	83.7	82.4	78.5	73.0	62.4	56.4
		MCDB	54.8	61.7	72.5	80.3	85.8	90.4	93.5	90.9	87.0	81.0	70.8	60.7
	2%	WB	44.9	49.7	59.9	65.0	72.8	78.4	81.4	80.0	76.3	68.4	58.6	50.0
		MCDB	49.2	56.3	65.6	74.8	81.7	87.6	91.3	88.1	85.1	75.7	65.3	52.4
	5%	WB	41.0	45.1	55.7	61.7	70.2	76.5	79.4	77.9	74.2	65.1	54.7	44.4
		MCDB	46.4	50.3	62.8	71.0	78.0	84.9	88.6	85.3	81.8	71.3	61.6	48.4
	10%	WB	37.0	40.6	50.3	58.7	67.5	74.5	77.5	75.7	71.8	61.3	50.3	39.9
		MCDB	41.3	45.5	57.8	67.1	74.4	82.2	85.7	82.5	79.0	68.2	56.9	44.1
Mean Daily Temperature Range	5% DB	MDBR	16.1	16.5	18.2	20.2	18.9	18.6	19.0	18.6	20.9	19.8	17.8	15.1
		MCDBR	22.1	24.0	26.4	26.8	22.5	20.9	21.2	21.8	23.5	24.0	24.7	20.3
		MCWBR	16.7	16.5	16.6	16.2	12.5	11.1	10.6	11.8	11.9	13.6	15.9	15.4
	5% WB	MCDBR	21.5	22.3	22.8	23.4	19.8	18.8	19.0	18.8	20.4	20.0	22.3	18.9
		MCWBR	16.6	16.4	16.1	15.9	12.2	11.8	11.6	12.3	11.5	13.8	16.1	15.6
Clear-Sky Solar Irradiance	taub	0.287	0.317	0.332	0.387	0.408	0.418	0.432	0.420	0.379	0.338	0.298	0.286	
	taud	2.433	2.348	2.397	2.259	2.268	2.282	2.237	2.292	2.354	2.467	2.514	2.457	
	Ebn at Noon	278	283	292	280	276	272	267	267	272	272	272	268	
	Edh at Noon	25	31	34	42	42	42	43	40	35	28	23	23	
All-Sky Solar Radiation	RadAvg	615	921	1196	1515	1784	1981	1995	1735	1477	994	678	525	
	RadStd	60	68	103	110	141	191	149	105	103	113	57	52	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page